

Explainable AI

CS 485/698

Credits: 3

Mode: Face-to-Face

Location: ME 221

Lecture: Tuesday/Friday, 2:30-3:50pm

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Office Hours: Tuesdays 1:00-2:00pm

Course Description

This course introduces technical methods for making machine learning models more transparent and understandable. Topics include intrinsically interpretable models, post hoc explanations (e.g., Shapley values, saliency maps), visualization of model internals (e.g., attention maps, neuron activations), surrogate modeling, mechanistic interpretability, and communication bottlenecks. Visualization is a central theme throughout the course, both as a practical tool and as a key research frontier.

The study of XAI is valuable for multiple reasons:

- **Deeper model understanding:** Students develop intuition about how neural networks process information internally
- **Critical thinking:** Learning to question and validate model outputs builds analytical skills
- **Communication skills:** Translating technical AI concepts for diverse audiences is increasingly valuable
- **Research preparation:** XAI is an active research frontier with opportunities for undergraduate and graduate research

Prerequisites

CS370 (Introduction to Artificial Intelligence) and CS375 (Introduction to Machine Learning) with a grade of C or better.

Alternative prerequisites for graduate students: CS670 (Artificial Intelligence) or CS675 (Machine Learning).

Background in calculus, linear algebra, probability, and Python programming is assumed.

Course Textbooks and Resources

Primary Resource:

- [*Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*](#) by Christoph Molnar (free online)

Supplementary Resources:

- Peer-reviewed articles from venues such as NeurIPS, ICML, ICLR
- Interactive explainers from distill.pub
- Tools and demonstrations from [PAIR AI explorables](https://PAIR.AI/explorables)
- "[Visualization for AI explainability \(visXAI\)](https://visx.ai)" workshop proceedings

[Note: Specific readings subject to instructor updates to reflect current research]

Learning Outcomes

This course emphasizes deep engagement with the methods, challenges, and communicative goals of explainable AI. By the end of the course, students will have developed the ability to:

- Analyze** and **compare** a range of explainable AI methods, including intrinsically interpretable models, post hoc techniques, and surrogate approaches, in terms of their assumptions, strengths, and limitations.
- Create** and **manipulate** visual explanations of model behavior, using methods such as saliency maps, attention visualization, and concept activation to surface internal representations.
- Evaluate** the faithfulness, robustness, and human-interpretability of model explanations across different contexts and audiences.
- Design** and **conduct** lightweight experiments to probe model internals or feature influence, using tools like knockout experiments, probing classifiers, and sensitivity analysis.
- Communicate** technical explanations of AI models and their outputs to both expert and non-expert audiences, through clear writing, visuals, and presentations.

Coursework and Assessment

Hands-On Assignments [20%]: Four assignments focused on active engagement with various interpretability methods. [Outcomes: b,c,e]

Projects [50%]: Four projects building from basic interpretability to advanced techniques:

- Project 1: Tabular Data Analysis (12.5%) [Outcomes: a,c,e]
- Project 2: Image Saliency Methods (12.5%) [Outcomes: a,b,c]
- Project 3: Language Model Interpretability (12.5%) [Outcomes: b,d,e]
- Project 4: Student Presentations (12.5%) [Outcomes: a,b,d,e]

All assignments and projects will be due Fridays 12:00 pm.

Midterm [10%]: Week 8 examination, covering input attribution and some internal structure.

[Outcomes: a,c,e]

Final Exam [20%]: Comprehensive examination covering theoretical foundations and practical applications. [Outcomes: a,c,e]

Late work policy: 48 free late hours. Additional lateness reduces the grade 2% per hour.

Delta between 698 and 485

Graduate students in the course will be treated more like researchers in XAI than the undergraduates.

- **Additional readings:** research paper assigned every week, with brief overviews delivered to the class periodically
- **Extended scope of projects**
- **Advanced tier of methods for final student presentations**
- **Higher expectation for critical analysis in all assignments**
- **Modified midterm, final exam**

Syllabus

Week	Topic	Primary data domain
1	What is XAI and why? Linear models	Tabular
2	Decision trees, ensembles, rule-based explanations	Tabular
3	Feature importance Shapley values	Tabular
4	Information theory	Tabular
5	Surrogate models/LIME Exemplars, influential instances, counterfactuals	Tabular/Image
6	Gradient-based methods Adversarial examples	Image
7	CNN feature visualization	Image
8	CLIP, Concept bottleneck models Midterm	Image/Text
9	Attention, transformers	Image/Text
10	Mechanistic interpretability	Text
11	Language-based explanations, chain of thought Intervening, jailbreaking	Text
12	Automated interpretability, evaluation Visualization	All
13	Current trends Open problems	All
14	Student presentations	

Course policies

Grade Corrections: Check the grades in course work and report errors promptly. Please try and resolve any issue within one week of the grade notification.

Incomplete: A grade of I (incomplete) is given in rare cases where work cannot be completed during the semester due to documented long-term illness or unexpected absence for other serious reasons. A student needs to be in good standing (i.e. passing the course before the absence) and receives a provisional I if there is no time to make up for the documented lost time; an email with a timeline of what is needed to be done will be sent to the student. Note that an I must always be resolved by the end of the next semester.

Collaboration and External Resources for Assignments: Some homework will be challenging. You are advised to first try and solve all the problems on your own. For problems that persist you are welcome to talk to the course assistant or the instructor. You are also allowed to collaborate with your classmates and search for solutions online. But you should use such solutions only if you understand them completely (admitting that you don't understand something is way better than copying things you don't understand). Also make sure to give the appropriate credit and citation.

Requesting Accommodations: If you need an accommodation due to a disability please contact Marsha Williams-Nicholas (marsha.williamsnicholas@njit.edu), Associate Director of the Office of Accessibility Resources and Services, to discuss your specific needs. A Letter of Accommodation Eligibility from the office authorizing student accommodations is required.

Generative AI Tools and Other External Resources: Sometimes you may come across code, text or other helpful information online, or you may be able to generate it using AI tools such as ChatGPT or other Large Language Models (LLMs). In most cases, you will be allowed to integrate this information into your solution. However, if you do, you must always give the appropriate credit and citations (e.g. links) for the material you use (especially when you use the code and text you found online). In the case you use an LLM, you must say that you did so, and present the entire transcript of your 'conversation' with it, which should show what you asked and how you guided it, or were guided by it to the delivered solution. Your 'conversation' with it must be entirely yours, and sufficiently different from that of other students. Failure to give appropriate credit when using the work of others (whether human or AI) is considered plagiarism, and may lead to disciplinary action under NJIT's Academic Integrity policy (see below).

Academic Integrity: Academic Integrity is the cornerstone of higher education and is central to the ideals of this course and the university. Cheating is strictly prohibited and devalues the degree that you are working on. As a member of the NJIT community, it is your responsibility to

protect your educational investment by knowing and following the academic code of integrity policy that is found at this [link](#).

Please note that it is my professional obligation and responsibility to report any academic misconduct to the Dean of Students Office. Any student found in violation of the code by cheating, plagiarizing or using any online software inappropriately will result in disciplinary action. This may include a failing grade of F, and/or suspension or dismissal from the university. If you have any questions about the code of Academic Integrity, please contact the Dean of Students Office at dos@njit.edu